

MIRAGE25: ACM MM25 Multimodal Interleaved Reasoning and Generation Challenge

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Abstract

We introduce the **MIRAGE Challenge**, a comprehensive benchmark for multimodal interleaved reasoning and generation, to ACM MM 2025. The challenge aims to evaluate models' abilities to both understand and generate content from complex, multimodal contexts consisting of interlinked images and text. The challenge is accompanied by the **MIRAGE Dataset**, comprising 263.7K high-quality instruction-response pairs across 35 tasks in two tracks: reasoning and generation. These pairs span 20 diverse scenarios, from surveillance to artistic creation, ensuring broad coverage. The challenge includes seven major categories: Multi-Image Reasoning, Document and Knowledge-Based Understanding, Interactive Multi-Modal Communication, Multi-Image Discrimination, Sequential Visual Generation, Material-based Image Coloring, and Visual Reference Customization. Hosting the MIRAGE Challenge at MM 2025 will drive significant progress in unified multimodal learning and inspire broad involvement in developing more versatile AI systems capable of both understanding and generating multimodal content. Challenge details and participation information are available at <https://mm25mirage.github.io/mirage/>.

CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning**; *Planning and scheduling*; **Computer vision**.

Keywords

Visual Story Generation, Visual Question Answering, Multi-Image Reasoning, Image Generation, Visual Reference Customization

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1 Challenge Overview

To facilitate research on interleaved vision-language instruction following capability, we build **MIRAGE**, a comprehensive challenge of **35 tasks of 7 categories** covering **20 diverse scenarios**. MIRAGE has three important properties: (1) **Mixed forms of reasoning and generation instruction**, ranging from visual reference customization needing pixel-level generation, to multi-image discrimination requiring fine-grained understanding. (2) **Interleaved vision-language contexts**, all instructions involve sequences of interconnected images and texts that fully express reasoning and generation tasks. (3) **Diverse range of instruction-following scenarios**, the benchmark covers various real-world scenarios.

This challenge consists of two sub-tracks: **Multimodal Interleaved Instruction Reasoning** and **Multimodal Interleaved Content Generation**. Specifically, we extensively gather a wide variety of multi-modal datasets from various tasks. Below is an introduction to each track:

Track A: Multimodal Interleaved Instruction Reasoning:

This track focuses on the understanding analysis and inference between multiple images. Participants are provided with interleaved image-text sequences and predefined open-ended questions or multi-choice options. It tests the model's reasoning ability to understand temporal progression, logical connections, and contextual relationships. It consists of the following subtasks:

- **Multi-Image Reasoning**. This category focuses on the comparative analysis between images and texts.
- **Document and Knowledge-Based Understanding**. This category requires extraction and reasoning of information from structured formats.
- **Interactive Multi-Modal Communication**. This category focuses on the dynamic interaction between visual and textual modalities in a conversational context.
- **Multi-Image Discrimination**. This task involves analyzing two given photos to determine their similarities and discrimination across multiple dimensions.

Track B: Multimodal Interleaved Content Generation: This track focuses on generating subsequent images and text descriptions (not necessary) followed by sequential interleaved image-text contexts and instructions. Participants are provided with interleaved image-text sequences and demand instructions for generating images. It consists of the following three categories of subtasks:

Table 1: Summary of the multimodal interleaved reasoning and generation tasks in MIRAGE challenge.

	Task	Scenario	Dataset	Metrics
Multimodal Interleaved Instruction Reasoning	Multi-Image Reasoning			
	Visual Change Captioning	Surveillance	Spot-the-Diff [14]	ROUGE-L
	Visual Change Captioning	Synthetic	CLEVR-Change [9]	ROUGE-L
	Visual Relationship Expressing	General	IEEdit [29]	ROUGE-L
	Subtle Difference Expressing	Fine-Grained	Birds-to-Words [6]	ROUGE-L
	Image-Set QA	Driving Recording	nuScenes [3]	Accuracy
	Industrial Inspection	Industrial	VISION [1]	Accuracy
	Fashion QA	Fashion	Fashion200K [8]	Accuracy
	Property Coherence	General	MIT-States-PropertyCoherence [12]	Accuracy
	State Transformation Coherence	General	MIT-States-StateCoherence [12]	Accuracy
	Visual Step Matching	Recipe	RecipeQA-ImageCoherence [31]	Accuracy
	Multi-Image Visual Entailment	General	NLVR2 [27]	Accuracy
	Ambiguity Analysis	Mobile Photo	VizWiz [2]	Accuracy
	Document and Knowledge-Based Understanding			
	Slide QA	Slide	SlideVQA [30]	Accuracy
	OCR QA	Book Cover	OCR-VQA [22]	Accuracy
	Document QA	Document Image	DocVQA [21]	Accuracy
	Webpage QA	Webpage	WebQA [4]	Accuracy
	Textbook QA	Textbook	TQA [15]	Accuracy
	Complex Multimodal QA	Wikipedia	MMQA [28]	Accuracy
	Interactive Multi-Modal Communication			
	Conversational Embodied Dialogue	Embodied	ALFRED [26]	ROUGE-L
	Multi-Modal Dialogue	Conversation	MMCoQA [17]	ROUGE-L
Multimodal Interleaved Content Generation	Multi-Image Discrimination			
	Facial Comparison	General	LFW [10]	Accuracy
	Similarity Dimension Selection	General	Totally-Looks-Like [24]	Accuracy
	Sequential Visual Generation			
	Animated Story Completion	Cartoon	AESOP [23]	ROUGE-L& Similarity
	Animated Story Completion	Cartoon	PororoSV [16]	ROUGE-L& Similarity
	Animated Story Completion	Cartoon	FlintstonesSV [7]	ROUGE-L& Similarity
	Sequential Photo Storytelling	Album	VIST [11]	ROUGE-L& Similarity
	Sequential Photo Storytelling	Cartoon	DiDeMeSV [19]	ROUGE-L& Similarity
	Comic Dialogue Identification	Cartoon	COMICS-Dialogue [13]	ROUGE-L& Similarity
	Comic Panel Identification	Cartoon	COMICS-Panel [13]	ROUGE-L& Similarity
	Recipe Completion	Recipe	RecipeQA-TextClose [31]	ROUGE-L& Similarity
	Visual Step Close	Recipe	RecipeQA-VisualClose [31]	ROUGE-L& Similarity
	Material-based Image Coloring			
	Material Transfer	Industrial	FMD [25]	Similarity
	Texture Transfer	Physical	KTH-TIPS2 [20]	Similarity
	Visual Reference Customization			
	Virtual Try-on	Fashion	VTON-HD [5]	Similarity
	Visual Reference	General	MSCOCO [18]	Similarity

- **Sequential Visual Generation.** This task category encompasses the subsequent image generation of coherent narratives based on sequential visual inputs.
- **Material-based Image Coloring.** The task involves coloring a specific object in the target image based on the given material image to get the corresponding material.
- **Visual Reference Customization.** This task requires the model to extract and apply pixel-level control from images.

A brief summary of MIRAGE is present in Table 1.

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